Oscillations of a Driven Quadruple Pendulum

Soham Bhar and Parth Bhargava
PC3236, Computational Methods in Physics
National University of Singapore
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The motion of four point masses on rigid rods, driven by a periodic torque at the top pivot and damped by a viscous torque on each rod, was computed from its Lagrangian equations of motion with a self-written fourth-order Runge-Kutta integrator, which solves the 4×4 mass-matrix system by LU decomposition at every stage. The integrator reproduced the linear normal-mode frequencies to within 4×10^{-6}, converged at fourth order, and satisfied the work-energy balance of the driven, damped motion to 2×10^{-10} of the energy supplied. For a drive of 8.0 N m at 0.8 Hz, the largest Lyapunov exponent, from eight 1000 s two-trajectory estimates, is (1.03 ± 0.02) s^{-1}. The motion is therefore chaotic, and an error grows tenfold every 2.2 s, so a trajectory computed with a 0.5 ms step follows the exact one to within 0.01 rad for only about 20 s; beyond that only averaged properties are meaningful. A sweep of the drive amplitude from 0 to 10 N m found periodic motion up to 6.00 N m, aperiodic motion with an exponent indistinguishable from zero between 6.05 and 6.50 N m, consistent with quasi-periodicity, and chaos from 6.55 N m, interrupted by a period-two window near 6.8 N m, with the exponent rising to 1.7 s^{-1} at 10 N m.

1. DESCRIPTION OF THE PROBLEM

A pendulum driven by a periodic torque and slowed by friction is the standard example of a deterministic system whose long-time motion can be irregular: for some drive strengths the motion settles onto a periodic orbit, for others it never repeats and two nearly identical starting states separate exponentially fast [1,2]. The rate of that separation, the largest Lyapunov exponent λ, sets the time beyond which a simulation can no longer predict the actual trajectory. A chain of four pendulums has four coupled degrees of freedom, so energy put into the top rod is passed down the chain through the nonlinear coupling, and the outer, lighter-loaded rods can be driven over the top.

The system studied is shown in Fig. 1: four point masses $m_1 = m_2 = m_3 = m_4 = 1.0$ kg on rigid, massless rods of lengths $L_1 = 0.40$ m, $L_2 = 0.35$ m, $L_3 = 0.30$ m and $L_4 = 0.25$ m, a drive of amplitude $\tau_0 = 8.0$ N m and angular frequency $\omega_d = 1.6\pi$ rad/s (period $T_d = 1.25$ s), and damping coefficient $b = 0.10$ N m s/rad. All runs start from $\boldsymbol{\theta}(0) = (0.4, 0.6, 0.3, 0.5)$ rad at rest unless stated. The project asks for the angles $\theta_i(t)$ from a self-written integrator. This report also asks three questions that the trajectories alone cannot answer: over what time the computed trajectory is accurate, whether the motion at the nominal parameters is chaotic, as measured by $\lambda > 0$ with a stated uncertainty, and how the long-time response changes with the drive amplitude τ_0.

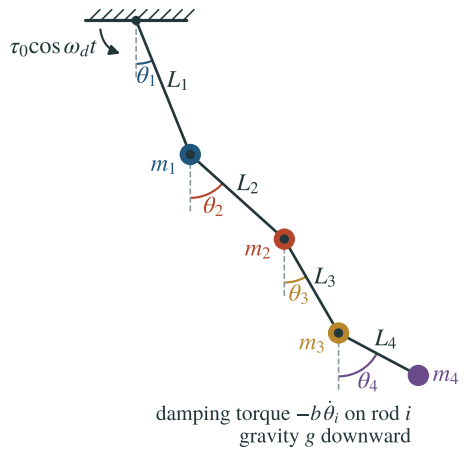


FIG. 1. The driven quadruple pendulum. Rod i has length L_i and carries a point mass m_i at its lower end; θ_i is its angle from the downward vertical, positive anticlockwise. A torque $\tau_0 \cos \omega_d t$ acts on rod 1 at the fixed pivot, and every rod experiences a damping torque $-b\dot{\theta}_i$. The angles drawn are illustrative.

2. EQUATIONS TO BE SOLVED

2.1 Lagrangian

With θ_i measured from the downward vertical (Fig. 1), the position of bob i is

$$x_i = \sum_{k=1}^{i} L_k \sin\theta_k, \quad y_i = -\sum_{k=1}^{i} L_k \cos\theta_k. \tag{1}$$

Bob i moves with the velocities of all rods $k \le i$, so the product $\dot{\theta}_j\dot{\theta}_k$ appears in the kinetic energy of every bob $i \ge \max(j,k)$. Summing over bobs,

$$T = \frac{1}{2}\sum_i m_i(\dot{x}_i^2 + \dot{y}_i^2) = \frac{1}{2}\sum_{j,k} C_{jk}\cos(\theta_j - \theta_k)\dot{\theta}_j\dot{\theta}_k, \tag{2}$$

$$V = \sum_i m_i g y_i = -g\sum_k \mu_k L_k \cos\theta_k, \tag{3}$$

where $\mu_k = \sum_{i \ge k} m_i$ (kg) is the mass carried below joint k and $C_{jk} = \mu_{\max(j,k)} L_j L_k$ (kg m^2).

2.2 Generalised Forces

The drive does virtual work $\tau\,\delta\theta_1$, so it enters only the equation for θ_1 as $Q_1^{\mathrm{drive}} = \tau_0 \cos\omega_d t$. The damping is modelled by the Rayleigh dissipation function $\mathcal{R} = \frac{1}{2} b \sum_i \dot\theta_i^2$, giving $Q_i^{\mathrm{damp}} = -\partial\mathcal{R} / \partial\dot\theta_i = -b\dot\theta_i$ [3]: a torque on each rod opposing its absolute angular velocity, as for a rod turning in still air. It is not joint friction, which would depend on $\dot\theta_i - \dot\theta_{i-1}$, nor drag on the bobs, which would couple all four equations.

2.3 Equations of Motion

The Euler-Lagrange equations $\frac{\mathrm{d}}{\mathrm{d}t}\big(\partial T / \partial\dot\theta_i\big) - \partial T / \partial\theta_i + \partial V / \partial\theta_i = Q_i$ with Eq. (2) and Eq. (3) use

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial\dot\theta_i} - \frac{\partial T}{\partial\theta_i} = \sum_j C_{ij}\big[\cos(\theta_i-\theta_j)\ddot\theta_j + \sin(\theta_i-\theta_j)\dot\theta_j^2\big],$$

in which the terms in $\dot\theta_i\dot\theta_j$ from the two derivatives cancel. The four equations take the matrix form

$$\boldsymbol{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} = \boldsymbol{F}(\boldsymbol{\theta},\dot{\boldsymbol{\theta}},t), \tag{4}$$

$$M_{ij} = C_{ij}\cos(\theta_i-\theta_j), \tag{5}$$

$$\begin{aligned} F_i = &-\sum_j C_{ij}\sin(\theta_i-\theta_j)\dot\theta_j^2 - \mu_i g L_i \sin\theta_i \\ &-b\dot\theta_i + \delta_{i1}\tau_0\cos\omega_d t, \end{aligned} \tag{6}$$

with $\boldsymbol{M}$ in kg m^2 and $\boldsymbol{F}$ in N m. The first term of F_i is the centrifugal coupling ($j = i$ contributes nothing), the second gravity, the third damping and the last the drive. Because T is positive for any nonzero velocity, $\boldsymbol{M}$ is symmetric positive definite and Eq. (4) always has a unique solution. With the state $\boldsymbol{y} = (\theta_1,\ldots,\theta_4,\omega_1,\ldots,\omega_4)$, $\omega_i = \dot\theta_i$, the system becomes eight first-order equations,

$$\dot\theta_i = \omega_i, \quad \dot\omega_i = \big[\boldsymbol{M}^{-1}\boldsymbol{F}\big]_i. \tag{7}$$

Multiplying Eq. (4) by $\dot{\boldsymbol{\theta}}$ gives the energy balance used as a test,

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \tau_0\cos(\omega_d t)\dot\theta_1 - b\sum_i\dot\theta_i^2, \quad E = T + V. \tag{8}$$

2.4 Small Oscillations

For small angles, Eq. (4) reduces to $\boldsymbol{C}\ddot{\boldsymbol{\theta}} = -\boldsymbol{K}\boldsymbol{\theta}$ with $\boldsymbol{K} = \mathrm{diag}(\mu_i g L_i)$. Its normal-mode angular frequencies, from $\boldsymbol{K}\boldsymbol{v} = \Omega^2\boldsymbol{C}\boldsymbol{v}$, are $\Omega = 3.09$, 7.65, 11.98 and 16.25 rad/s. They differ from the frequencies $\sqrt{g / L_i} = 4.95$–6.26 rad/s of isolated pendulums because the coupling is strong. The drive frequency $\omega_d = 5.03$ rad/s lies between the two lowest modes and is not resonant with either. Adding damping, the slowest mode decays with time constant 14.6 s, which sets how long transients must be discarded.

2.5 Runge-Kutta Integration

Eq. (7) is advanced with the classical fourth-order Runge-Kutta (RK4) method [4,5]. For $\dot{\boldsymbol{y}} = \boldsymbol{f}(t,\boldsymbol{y})$ and step h,

$$\begin{aligned} &\boldsymbol{k}_1 = \boldsymbol{f}(t_n,\boldsymbol{y}_n), \quad \boldsymbol{k}_2 = \boldsymbol{f}\Big(t_n+\tfrac{h}{2},\boldsymbol{y}_n+\tfrac{h}{2}\boldsymbol{k}_1\Big), \\ &\boldsymbol{k}_3 = \boldsymbol{f}\Big(t_n+\tfrac{h}{2},\boldsymbol{y}_n+\tfrac{h}{2}\boldsymbol{k}_2\Big), \quad \boldsymbol{k}_4 = \boldsymbol{f}(t_n+h,\boldsymbol{y}_n+h\boldsymbol{k}_3), \\ &\boldsymbol{y}_{n+1} = \boldsymbol{y}_n + \frac{h}{6}(\boldsymbol{k}_1+2\boldsymbol{k}_2+2\boldsymbol{k}_3+\boldsymbol{k}_4). \end{aligned} \tag{9}$$

The local error is $O\big(h^5\big)$ and the global error over a fixed interval $O\big(h^4\big)$. Each evaluation of $\boldsymbol{f}$ requires the solution of Eq. (4) for $\ddot{\boldsymbol{\theta}}$.

2.6 Diagnostics of Chaos

The largest Lyapunov exponent is

$$\lambda = \lim_{t\to\infty}\frac{1}{t}\ln\frac{|\delta\boldsymbol{y}(t)|}{|\delta\boldsymbol{y}(0)|}, \tag{10}$$

the mean exponential growth rate (s^{-1}) of an infinitesimal separation $\delta\boldsymbol{y}$ between two trajectories [1]. A bounded motion with $\lambda > 0$ is chaotic; a stable periodic orbit has $\lambda < 0$. The limit does not depend on the norm, here the Euclidean norm of the eight state components in rad and rad/s. The stroboscopic Poincaré section records the state once per drive period, at $t = nT_d$: a period-p orbit gives p repeating points, and a chaotic orbit gives a set that never repeats [2].

3. DESCRIPTION OF COMPUTATIONAL METHODOLOGY AND IMPLEMENTATION

3.1 Structure of the Code

The parameters, the right-hand side of Eq. (7) and the integrator are defined first; the tests and analyses (Sec. 3.3, Sec. 3.4) call them, write every quoted number to one results file and draw the figures. The physics functions accept arrays of states of shape $(N, 8)$, so that N trajectories (different initial states or drive amplitudes) advance together in one RK4 step. Table I links the code to the equations.

3.2 Choice of Syntax

The mass matrix and forcing vector are written as array expressions, Eq. (5) and Eq. (6) term for term, rather than as ten matrix elements and four rows typed out by hand. The explicit form was used first; it gives the same accelerations to within 10^{-14} (relative) for random states, but its twelve hand-typed coupling terms are where the first error of Sec. 3.6 occurred, and the array form cannot have a sign that differs between rows. Indexing with `[..., i]` lets the same functions act on one state or on a batch, and `np.linalg.solve` accepts a stack of 4×4 systems.

The accelerations are obtained with `np.linalg.solve`, which factors $\boldsymbol{M} = \boldsymbol{P}\boldsymbol{L}\boldsymbol{U}$ by Gaussian elimination with partial pivoting

TABLE I. Lines of the simulation code and the equations they implement.

Lines	Equation or operation
43–60	Parameters g, L_i, m_i, τ_0, ω_d, b, Δt; `MU` $= \mu_k$ by a reversed cumulative sum; `C` $= C_{jk}$ (Eq. (2)) from the outer product $L_j L_k$ and $\mu_{\max(j,k)}$; `GRAV` $= \mu_i g L_i$.
65–68	`mass_matrix`: Eq. (5). The array `d` holds all differences $\theta_i - \theta_j$; `C * np.cos(d)` forms the 4×4 matrix in one operation.
71–80	`forcing`: Eq. (6). `(C * np.sin(d)) @ om**2` is the sum $\sum_j C_{ij} \sin(\theta_i - \theta_j)\dot{\theta}_j^2$; gravity and damping are element-wise; the drive is added to row 1 only.
83–88	`derivs`: Eq. (7). `np.linalg.solve` solves Eq. (4) for $\ddot{\boldsymbol{\theta}}$ by LU decomposition; the result is joined to $\boldsymbol{\omega}$.
93–99	`rk4_step`: the four stages and weighted sum of Eq. (9).
102–115	`integrate_rk4`: the loop $t_n = nh$, $n = 0, \ldots, t_{\text{end}} / h$.
120–136	`bob_positions`, `kinetic_energy`, `potential_energy`: Eq. (1), T from the Cartesian bob velocities, and Eq. (3).
139–144	`derivs_with_work`: appends $\mathrm{d}W_d / \mathrm{d}t = \tau\dot{\theta}_1$ and $\mathrm{d}W_b / \mathrm{d}t = b \sum \dot{\theta}_i^2$ to the state, for Eq. (8).
169–208	`normal_mode_test`: $\boldsymbol{K}\boldsymbol{v} = \Omega^2 \boldsymbol{C}\boldsymbol{v}$ and the measured mode frequencies.
331–363	`benettin`: Eq. (10) by the two-trajectory method with renormalisation every T_d.
432–480	`bifurcation_scan`: the sweep of τ_0, period detection and $\lambda(\tau_0)$.

and then performs one forward and one back substitution [5]. Forming $\boldsymbol{M}^{-1}$ and multiplying by $\boldsymbol{F}$ would need the same factorisation plus four substitutions to build the inverse, and is not more accurate. Accuracy is in any case not limited by this step: over 10^5 random configurations the condition number of $\boldsymbol{M}$ lies between 15 and 46, the largest value occurring when all rods are aligned, so the solve loses at most two of the sixteen significant digits.

Times are generated as $t_n = nh$ rather than by repeated addition, and h divides T_d exactly ($T_d / h = 2500$ for $h = 5 \times 10^{-4}$ s). The Poincaré samples are then the states at every 2500th step, exactly in phase with the drive.

3.3 Tests of the Integrator

Four tests compare the computed motion with results known independently of the integrator: the linear normal modes, conservation of energy, the energy balance of the driven motion, and the convergence with step size.

3.3.1 Small oscillations

Started in each linear normal mode with a largest angle of 10^{-3} rad, with no drive or damping, the full nonlinear integrator oscillated at frequencies (from 20 periods of zero crossings) that agree with the predicted Ω to within 4×10^{-6} (relative). This tests $\boldsymbol{M}$, the gravity terms and the coupling between rods together; the coupled frequencies differ from the isolated-pendulum values $\sqrt{g / L_i}$ by up to 60%, so the agreement could not arise if the coupling terms were wrong.

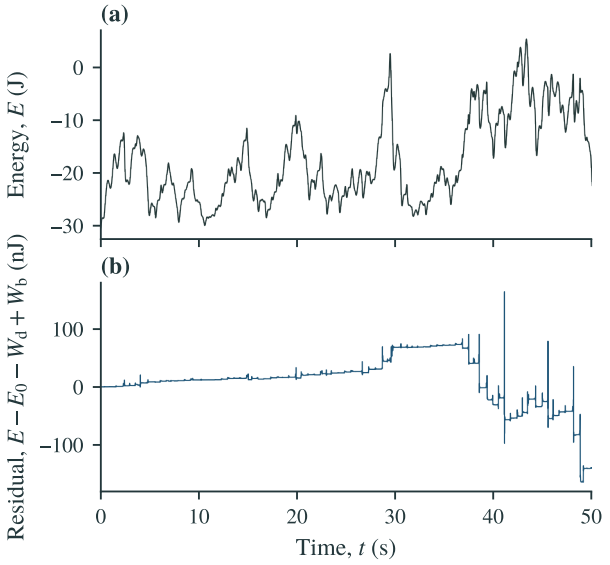


FIG. 2. Energy of the driven, damped pendulum at the nominal parameters. (a) Total mechanical energy E. (b) Residual of the energy balance, Eq. (8), with W_d the work done by the drive and W_b the energy dissipated, both integrated with the state.

3.3.2 Energy

With $\tau_0 = 0$ and $b = 0$, Eq. (8) requires E to be constant. Over 40 s from the reference initial state, the largest deviation of E was 5.1×10^{-8}, 1.7×10^{-9} and 5.6×10^{-11} J for $h = 2$, 1 and 0.5 ms, falling as $h^{4.9}$, consistent with fourth-order convergence; the motion has 3.6 J above the rest state. The kinetic energy is computed from the Cartesian bob velocities, independently of $\boldsymbol{M}$, so this also tests Eq. (5).

For the driven, damped motion E is not conserved, and the relevant test is Eq. (8) itself. Integrating the drive work W_d and the dissipated energy W_b alongside the state, the residual $E(t) - E(0) - W_d + W_b$ stayed below 1.7×10^{-7} J over the 50 s run (Fig. 2), while the drive supplied 704 J and damping removed 696 J; the balance holds to a relative precision of 2×10^{-10}.

3.3.3 Step size and accuracy over time

Fig. 3(a) shows the global error at $t = 5$ s against the step size, relative to a run with $h = 31.25$ µs. Between successive halvings the error falls by factors that approach $2^4 = 16$ (observed orders 1.4, 3.5, 3.8, 3.9 and 4.0), so the step is in the asymptotic fourth-order range; the error at the working step $h = 5 \times 10^{-4}$ s is 1.6×10^{-7} rad.

Because the motion is chaotic, a small error does not stay small. Fig. 3(b) gives the error of the working trajectory against time by two independent estimates: Richardson extrapolation, 16 / 15 times the difference between runs with h and $h / 2$, and the difference from an adaptive Runge-Kutta 4(5) solution (scipy `solve_ivp`, relative tolerance 10^{-10}, absolute 10^{-12}). The two agree throughout, so the adaptive solution is the more accurate one and the difference measures the RK4 error. The error is 1.1×10^{-10} rad at 1 s, grows on average at 0.62 s^{-1}, and exceeds 0.01 rad at $t = 20.0$ s and 0.1 rad at 21.1 s. A change of 10^{-15} rad in $\theta_1(0)$, the size of a rounding error, grows to more

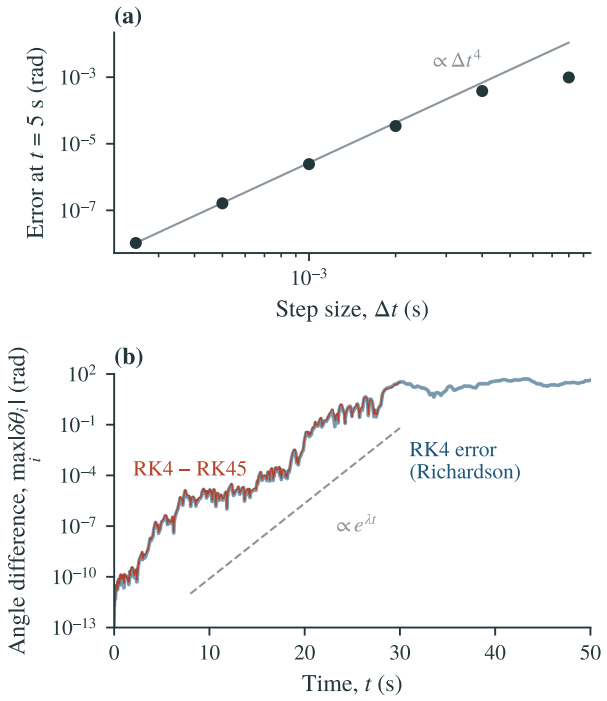


FIG. 3. Accuracy of the RK4 solution. (a) Largest angle error at t = 5 s against step size; the grey line has slope 4. (b) Error of the $h = 5 \times 10^{-4}$ s trajectory against time, estimated from a run with h / 2 (Richardson extrapolation, blue) and from the difference to an adaptive Runge-Kutta 4(5) solution with relative tolerance 10^{-10} (red). The dashed line shows growth at the rate λ found in Sec. 4.2.

than 0.01 rad by 28 s. The computed trajectory is therefore a quantitative prediction for about the first 20 s only; later it is one trajectory on the same attractor, which still serves for the statistical quantities (λ, the Poincaré section) that do not depend on following one particular trajectory.

3.4 Lyapunov Exponent and Amplitude Sweep

The exponent Eq. (10) was computed by the two-trajectory method of Benettin et al. [6]. A partner trajectory starts a distance $d_0 = 10^{-8}$ from the reference, in a random direction of the eight-dimensional state space. After every drive period the separation d is measured, $\ln(d\,/\,d_0)$ is added to a running sum, and the partner is moved back to distance d_0 along the current separation, so that the separation stays in the linear regime and aligns with the most unstable direction. The running estimate is the sum divided by the elapsed time. Eight reference trajectories, started from the reference state with $\theta_1(0)$ shifted by 0, 0.05, …, 0.35 rad, were first run for 150 s (about 10 decay times of the slowest mode) and then averaged over 1000 s each. The quoted uncertainty is the standard error of the mean of the eight values. Halving the step, raising d_0 to 10^{-6} and using only the second half of the averaging time test the numerical choices.

The amplitude sweep integrated 201 values of τ_0 from 0 to 10 N m in steps of 0.05 N m, all from the reference initial state, with $h = 10^{-3}$ s. After a transient of 150 s, 200 states were recorded at $t = nT_d$ over 250 s, together with λ by the method above. A response was classed as periodic with period $p \le 10$ if, over the last 40 samples, every state repeated after p samples to within 10^{-4} in $\cos\theta_i$, $\sin\theta_i$ and ω_i (rad/s).

3.5 Efficiency

One evaluation of Eq. (7) takes about 9 µs, so the 50 s reference run (10^5 steps, 4×10^5 linear solves) takes 4 s. Advancing many trajectories in one array spreads the interpreter overhead: 402 states take about 0.22 ms per evaluation, roughly 20 times less per state, which made the 1000 s Lyapunov runs and the 201-amplitude sweep (330 s) practical. The full analysis takes 13 min.

3.6 Debugging and Error Analysis

Two errors in earlier versions of the code changed the physics rather than the accuracy; both can be reproduced with the present code by a change of one parameter.

3.6.1 Sign of the centrifugal coupling

In the first version of the forcing vector, every coupling term $C_{ij}\sin(\theta_i - \theta_j)\dot\theta_j^2$ appeared with a positive sign on the right of Eq. (4). The error showed in the animation as the outer rods whipping ever faster, and in the printed maximum angular velocity. Re-deriving the Euler-Lagrange equation for a double pendulum placed the term on the left-hand side with a positive sign, hence with a negative sign in $\boldsymbol{F}$, as in Eq. (6). The error can be reproduced by reversing the sign of the term. Without drive or damping, the energy then changed by 1.3 J within 5 s, against 5.6×10^{-11} J in 40 s for the corrected code. With the drive and $b = 0$ the angular velocities reached 1850 rad/s within 30 s, the whipping described above; with $b = 0.10$ N m s/rad they stayed below 29 rad/s, close to the 31 rad/s of the corrected code. The undriven energy test, not the size of the velocities, is therefore the check that distinguishes the two signs.

3.6.2 Unbounded energy without damping

An earlier version had the drive but no damping. The energy then grew without limit, from −30.7 J at the start to 728 J after 200 s, with angular velocities up to 210 rad/s, about 0.1 rad per step at $h = 5 \times 10^{-4}$ s; the values printed after each run showed the growth. An unbounded energy has no steady state to analyse, and the step that is accurate for the bounded motion becomes too coarse. The viscous damping of Eq. (6) provides the energy sink: with $b = 0.10$ N m s/rad the mean dissipated power over 20–50 s, 15.9 W, nearly balanced the mean drive power, 15.6 W, the difference being the fall of E over that interval.

4. PRESENTATION OF RESULTS AND GRAPHS

4.1 Reference Trajectory

Over 50 s, rods 1 and 2 swung without turning over, with $|\theta_1| \le 1.87$ rad and $|\theta_2| \le 2.34$ rad, while rod 3 turned over once, near 46 s, and rod 4, the outermost and least massive, made 7.3 net revolutions (Fig. 4). The energy (Fig. 2(a)) rose from its initial value, 3.6 J above the rest state, and fluctuated

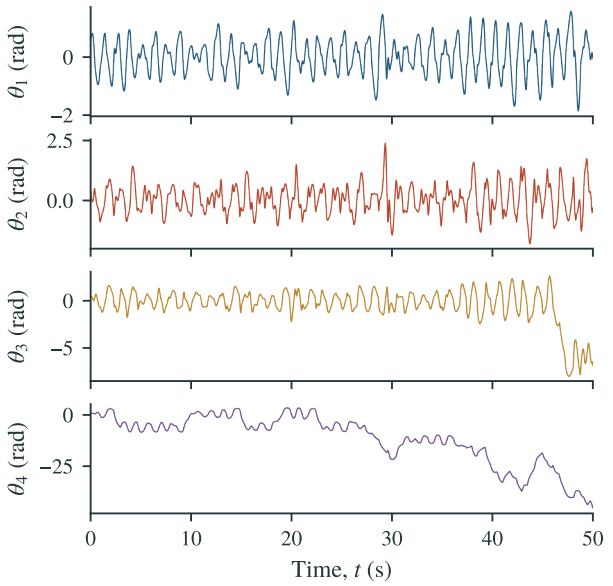


FIG. 4. Angles of the four rods, without wrapping, for the nominal parameters and reference initial state. Beyond about 20 s the curve is one trajectory on the attractor rather than an accurate prediction (Fig. 3).

irregularly between −30.7 and 5.2 J. Neither the angles nor the energy repeat over the run, but 50 s, of which only the first 20 s are accurate, cannot by itself distinguish chaos from a long transient or a long period; the following sections answer that question.

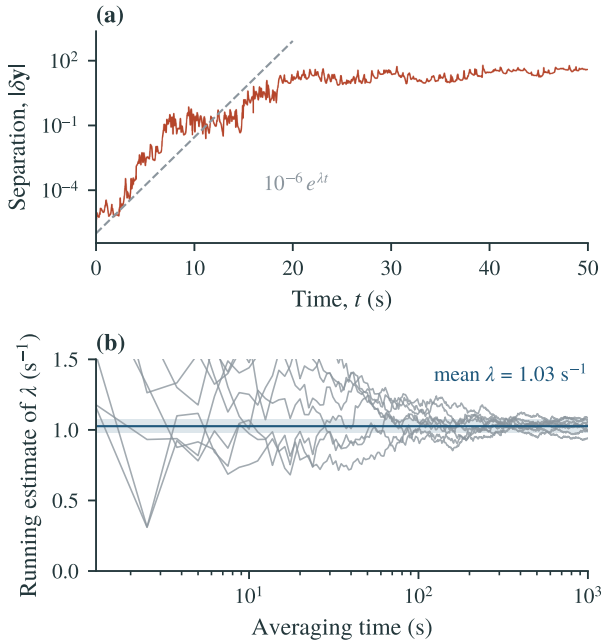


FIG. 5. Sensitivity to the initial state. (a) Distance in state space between the reference trajectory and one started with $\theta_1(0)$ larger by 10^{-6} rad; the dashed line is $10^{-6}e^{\lambda t}$ with the value of λ from (b). (b) Running estimate of the largest Lyapunov exponent for eight trajectories (grey); the blue line and band are their mean and standard deviation at the end of the run.

4.2 Separation of Nearby Trajectories

A trajectory started 10^{-6} rad away in θ_1 separated from the reference by 1.7×10^{-3} at 5 s and 0.17 at 10 s, and the separation reached order 1 at 15.0 s (Fig. 5(a)), after which it is limited by the size of the attractor. The growth fluctuates about an exponential because the local stretching rate varies along the trajectory.

The Benettin estimates converged after a few hundred seconds (Fig. 5(b)). The eight values range from 0.94 to 1.09 s^{-1}, with mean

$$\lambda = (1.03 \pm 0.02)\,s^{-1},$$

corresponding to a Lyapunov time $1 / \lambda \approx 0.97$ s, in which an error grows by a factor e. The second half of the averaging time alone gives 1.02 s^{-1}. Over 300 s, halving the step changed the mean from 1.03 to 1.01 s^{-1} and raising d_0 to 10^{-6} gave 1.03 s^{-1} these shifts are within the scatter expected from a shorter average, so the step and d_0 do not bias the result at the level of its uncertainty. The exponent is positive by many standard errors, which together with the bounded energy establishes that the motion at the nominal parameters is chaotic. The error growth of the integrator, 0.62 s^{-1} over 5–20 s (Fig. 3(b)), is of the same order: an integration error acts as a small perturbation of the initial state.

4.3 Poincaré Section

Fig. 6 shows the state at $t = nT_d$ for the eight trajectories after their transients, 6400 points. The points do not collapse onto a finite set, as a periodic orbit would, but fill bounded regions. The angles of rods 1 to 3 are concentrated near $\theta = 0$, with a sparse spread from the rare turns of rod 3, while the angle of rod 4 covers the whole circle because it turns over often. The eight trajectories, started from different states, fill the same regions, consistent with a single attractor. The section

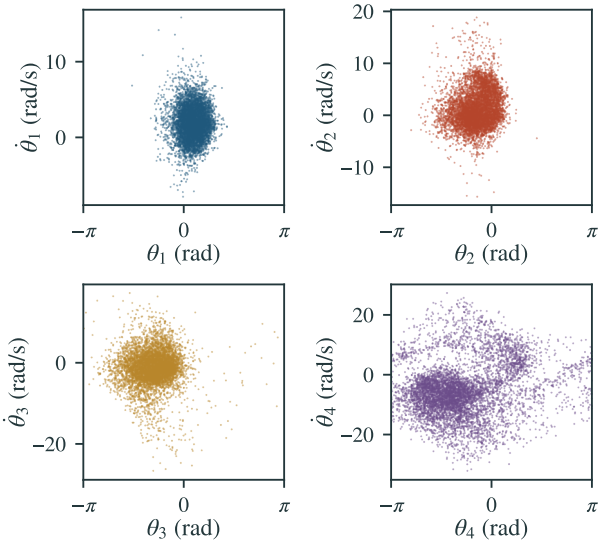


FIG. 6. Stroboscopic Poincaré section at the nominal parameters: angle (wrapped to $[-\pi, \pi)$) and angular velocity of each rod at $t = nT_d$, from the eight reference trajectories of Fig. 5(b), 6400 points in total.

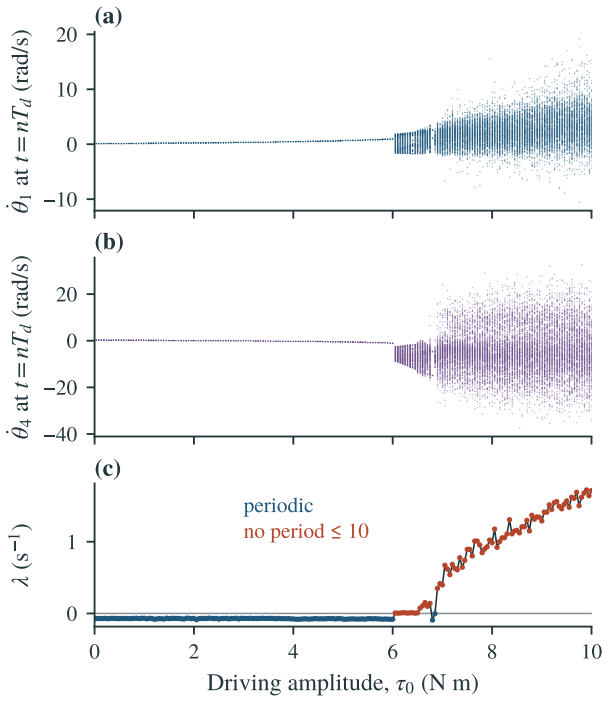


FIG. 7. Response against drive amplitude, every run from the reference initial state. (a, b) Angular velocities of rods 1 and 4 at $t = nT_d$, 200 samples per amplitude after a 150 s transient. (c) Largest Lyapunov exponent; blue points mark responses that repeat with a period of at most 10 drive periods, red points those that do not.

is a two-dimensional projection of a seven-dimensional set, so it shows no fine fractal structure and is not itself evidence of chaos; that evidence is $\lambda > 0$.

4.4 Dependence on the Drive Amplitude

For $\tau_0 \le 6.00$ N m every response repeated once per drive period ($p = 1$), with $\lambda = -0.073$ s^{-1} on average (Fig. 7(c)). This equals, within the averaging scatter, the decay rate $-1 / (14.6\ \text{s}) = -0.069$ s^{-1} of the slowest damped linear mode: a perturbation of the periodic orbit relaxes back as a small oscillation would. The period-1 orbit lost stability between 6.00 and 6.05 N m. From 6.05 to 6.50 N m no period up to 10 was found, yet $|\lambda|$ stayed below 0.011 s^{-1}, so nearby states did not separate exponentially. This is consistent with quasi-periodic motion, in which a second frequency incommensurate with ω_d appears, or with a period longer than $10T_d$. A clearly positive exponent first appeared at 6.55 N m ($\lambda = 0.07$–0.15 s^{-1} up to 6.75 N m). A period-2 window followed at 6.80 and 6.85 N m, and from 6.90 N m every response was aperiodic, with λ rising from 0.35 s^{-1} to about 1.7 s^{-1} at 10 N m. In panels (a) and (b) the same sequence appears as a single branch, a widening band and then a broad scatter that grows with τ_0.

The scatter of a single 250 s estimate of λ, from the root-mean-square difference between its full and second-half values in the chaotic range, is 0.08 s^{-1}. At 8.00 N m the sweep, with $h = 1$ ms, gave 0.99 s^{-1}, consistent with the reference value at $h = 0.5$ ms. The nominal amplitude therefore lies well inside the chaotic range rather than near its edge.

5. REFLECTIONS AND DISCUSSION

The Lyapunov exponent sets the useful length of any single computed trajectory. With $\lambda = 1.03$ s^{-1} an error grows tenfold every $\ln 10 / \lambda = 2.2$ s. The RK4 error after 1 s is about 10^{-10} rad, so it reaches 0.01 rad at 20 s, and even a rounding error of 10^{-15} rad does so by 28 s. Halving the step lowers the error by a factor 16 and so extends the accurate interval by only $\ln 16 / \lambda \approx 2.7$ s. Beyond about 20 s the trajectory in Fig. 4 is therefore one representative trajectory, and a comparison with another integrator, such as the adaptive RK4(5) solution, is informative only before the two diverge; their agreement to 1.4×10^{-10} rad over the first second and the same error growth thereafter show that both resolve the same equations. Quantities averaged over the attractor do not suffer from this limit: the eight Lyapunov estimates from different starting states lie between 0.94 and 1.09 s^{-1}, and halving the step changed their mean by less than its uncertainty.

The drive acts on the inner rod at a frequency between the two lowest normal modes, so at small amplitudes the chain follows it in a forced periodic oscillation. As the amplitude grows, the energy passed down the chain is enough to take the outer rod, which carries only m_4 and has the smallest moment of inertia $m_4 L_4^2$, over the top; once rods turn over, the coupling terms in Eq. (6) change sign irregularly. The sweep shows that this onset does not occur at a single threshold: periodic motion gives way first to motion with $\lambda \approx 0$, then to weak chaos, a periodic window, and fully developed chaos whose exponent keeps growing with τ_0.

The main limitations are as follows. The thresholds 6.05, 6.55 and 6.90 N m belong to this damping model, a torque on each rod proportional to its absolute angular velocity; joint friction or drag on the bobs would remove energy differently and would move them by an amount not assessed here. Every run of the sweep started from the same state, so coexisting attractors, which in driven oscillators often cause hysteresis, could make the thresholds depend on the initial state; only at 8.0 N m did eight different starting states lead to the same λ. The amplitude step of 0.05 N m resolves the period-2 window with two points and would miss narrower windows. The classification of the band from 6.05 to 6.50 N m as quasi-periodic rests on $\lambda \approx 0$ and on the absence of a period up to 10, not on a direct measurement of a second frequency. The 1 ms step of the sweep was checked only at 8.0 N m, although the angular velocities grow with τ_0. Finally, the two-dimensional projections of the Poincaré section cannot show the structure of a seven-dimensional attractor, and its dimension would require the full Lyapunov spectrum.

6. CONCLUSION

The Lagrangian equations of motion of the driven, damped quadruple pendulum were integrated with a self-written RK4 scheme and an LU solution of the mass-matrix system at every stage. The integrator reproduced the linear normal-mode frequencies to within 4×10^{-6}, converged at fourth order, conserved energy to 6×10^{-11} J over 40 s without drive or damping, and satisfied the work-energy balance of the driven, damped motion to 2×10^{-10} of the energy supplied. At $\tau_0 =$ 8.0 N m the largest Lyapunov exponent is (1.03 ± 0.02) s^{-1}

(standard error of eight 1000 s estimates), so the motion is chaotic and the computed trajectory with $h = 0.5$ ms is accurate to 0.01 rad for only the first 20 s. With increasing drive amplitude the response is periodic up to 6.00 N m, aperiodic with $\lambda \approx 0$ from 6.05 to 6.50 N m, chaotic from 6.55 N m apart from a period-2 window at 6.80–6.85 N m, with λ reaching 1.7 s^{-1} at 10 N m.

Specific improvements follow from these limitations. The nature of the band with $\lambda \approx 0$ could be settled by the frequency spectrum of the stroboscopic sequence or by the second Lyapunov exponent, computed with several partner trajectories and Gram-Schmidt orthogonalisation [6]. Sweeping τ_0 up and down with each run started from the final state of the previous one, and from several initial states, would reveal any coexisting attractors. A finer step of τ_0 between 6 and 7 N m, with averages longer than 250 s, would resolve the windows and reduce the scatter of $\lambda(\tau_0)$ below 0.08 s^{-1}. Repeating the sweep with joint friction would show how strongly the thresholds depend on the damping model. A smaller step is not an improvement for single trajectories, since each halving extends the accurate interval by less than 3 s.

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